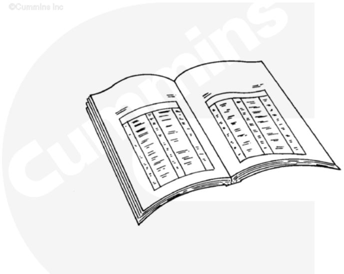
Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Remove all of the coolant and heater hoses. Refer to the OEM service manual.
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/41/41-010-033-tr.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/41/41-011-007-tr.html)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/41/41-006-026.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.

(/qs3/pubsys2/xml/en/procedures/100/100-003-011.html)

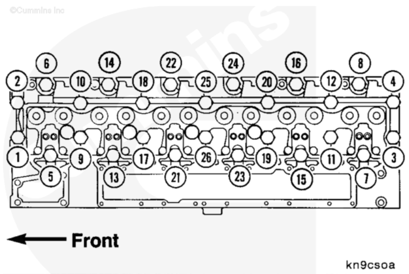
- Remove the rocker lever assemblies. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-008.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-014.html)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html)
- Remove the fan pulley and fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-036.html)

Note : In some applications, it can be easier to remove the thermostat housing to gain access to the exhaust manifold capscrews on cylinder number 1.

- Remove the thermostat housing. Refer to Procedure 008-014 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-014.html)
- Disconnect the coolant vent lines. Refer to Procedure 008-017 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-017.html)
- Remove the intake manifold cover and intake heater, if equipped. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/41/41-010-023.html)

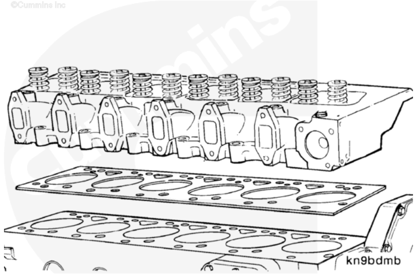
Remove

Remove the cylinder head capscrews in the sequence shown.



⚠ WARNING ⚠

The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



⚠ CAUTION ⚠

Do not lay the cylinder head on the combustion deck. This can damage the combustion deck.

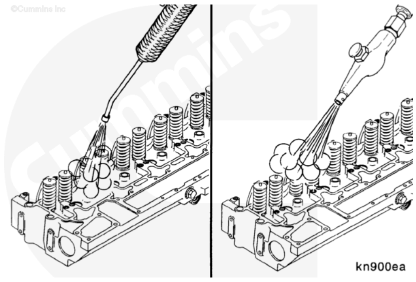
Use a hoist or hydraulic arm to remove the cylinder head. Make sure the head is removed in a direct upward motion.

Remove the cylinder head gasket from the cylinder block.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

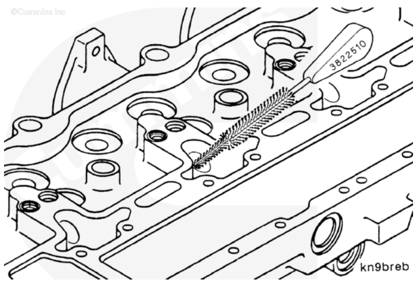
Steam-clean the cylinder head.

Note : Make sure to blow out all the capscrew holes.

Dry the cylinder head with compressed air.

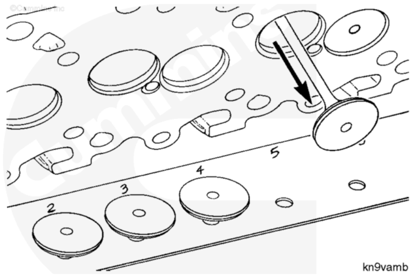
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

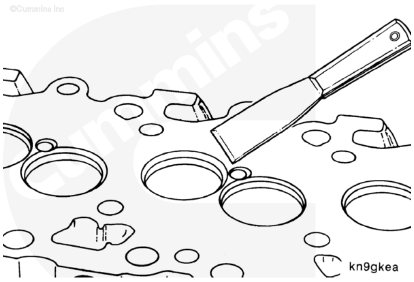


Clean the injector bores with an Cummins® injector bore brush, Part Number 3822510, and solvent.

Remove the valves from the cylinder head. Refer to Procedure 002-020 (/qs3/pubsys2/xml/en/procedures/41/41-002-020.html).

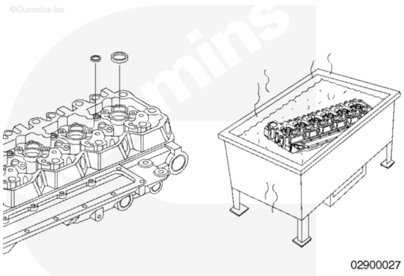


Scrape the gasket material from all gasket surfaces on the block and head.



Clean the buildup of deposits from the coolant passages.

Inspect the area within 1/8-inch of the combustion seal ring diameter. Any wear that can be felt with a fingernail within the 1/8-inch area is unacceptable, making the cylinder head **not** reusable.Wear beyond this 1/8-inch area will have no effect on future combustion sealing and the usability of the cylinder head.



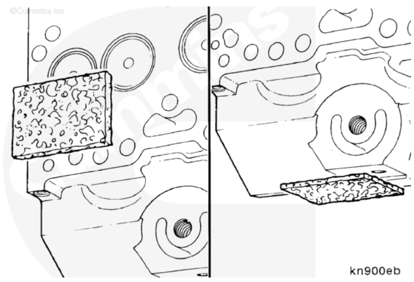
Clean carbon deposits from the valve pockets with a high quality steel wire wheel installed in a drill or a die grinder.

Note : An inferior quality wire wheel will lose steel bristles during operation, causing additional contamination.

Note : Excessive deposits can be cleaned in an acid tank. The expansion plugs must be removed before putting the cylinder head into an acid tank.

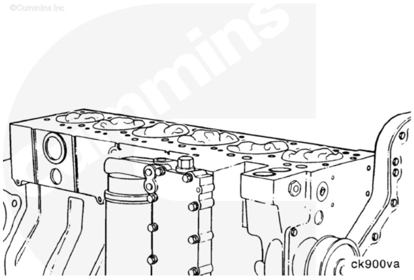
⚠ WARNING ⚠

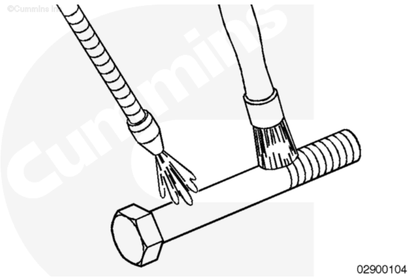
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the cylinder head combustion deck, exhaust manifold gasket surface, and valve cover gasket surface with Scotch-Brite™ 7448, Part Number 3823258, and diesel fuel or solvent.

Clean the cylinder block head deck surface. Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/41/41-001-026.html)





⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

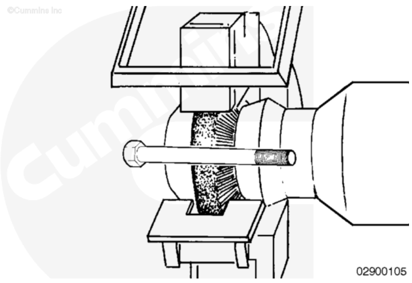
Cylinder Head Capscrews

⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews. Use of these solutions can cause corrosion and damage to the cylinder head capscrews.

Use a petroleum-based solvent to clean the cylinder head capscrews.

Clean the capscrews thoroughly with a wire brush, a soft wire wheel, or use a non-abrasive bead blast to remove deposits from the shank and threads of the cylinder head capscrews.

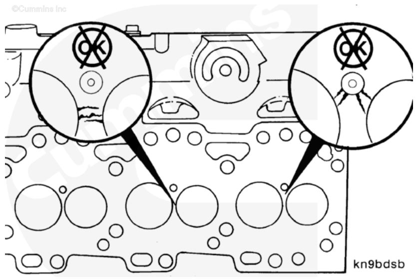


Cylinder Head

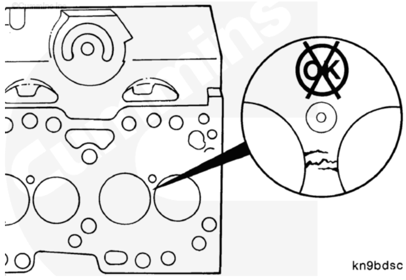
Inspect the cylinder head and valve seats for obvious damage that would prohibit reuse.

Inspect for cracks and damage to the deck surface that would result in loss of sealing.

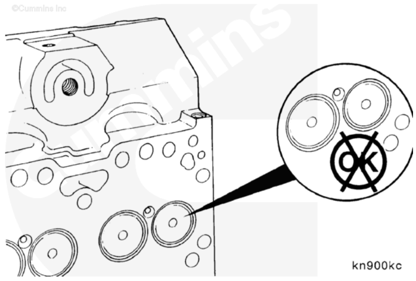
Note : If a crack in the cylinder head is suspected, pressure-test the cylinder head.



Cracks between the valve seats are **not** acceptable and the cylinder head **must** be replaced.

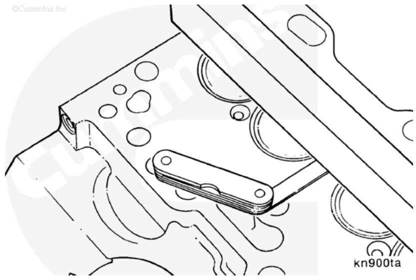


Inspect the valves for indications of leakage or burning.

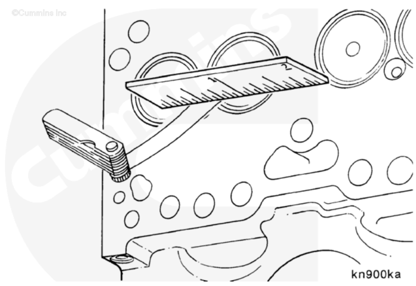


Use a straightedge and feeler gauge to check the cylinder head combustion surface for flatness.

Cylinder Head Flatness Specifications				
	mm		in	
End-to-End		0.20	MAX	0.008
Side-to-Side		0.076	MAX	0.003

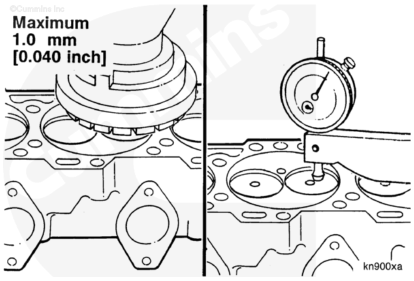


Use a small 51 mm [2 in] straightedge and a 0.0254-mm [0.001-in] feeler gauge to measure local flatness. Check between the cylinder bores and between the coolant passages. If the 0.001-inch feeler gauge fits under the 2-inch straightedge, the cylinder head **must** be replaced or resurfaced.



Note : A maximum of 1 mm [0.040 in] can be machined from the combustion surface of the cylinder head.

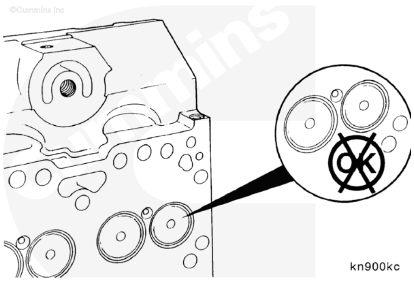
If the cylinder head is machined, place the valve in the respective bore and check the valve depth dimension. If the depth is less than the minimum specification, the valve seat will require machining. Refer to the Shop Manual, C Series Engines, Bulletin 3666008.



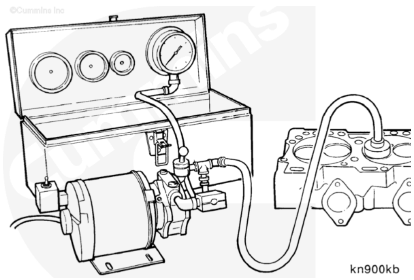
Valve Recess in Cylinder Head			
	mm		in
Exhaust	1.09	MIN	0.0430
	1.62	MAX	0.064
Intake	0.59	MIN	0.023
	1.12	MAX	0.044

Valve and Seat Leakage Test

Inspect the valves for indications of leakage or burning.

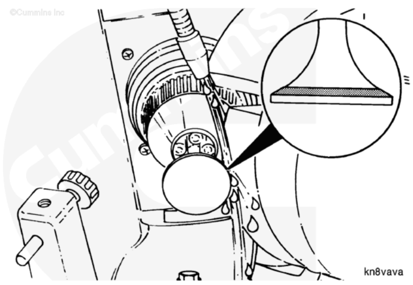


Note : If a leaking valve is suspected, vacuum test the valves and valve seats. The vacuum must not drop more than 25.4 mm Hg [1.0 in Hg] in 5 seconds.



Valve to Valve Seat Vacuum			
	mm-hg		in-hg
Used	457	MIN	18
New	635	MIN	25

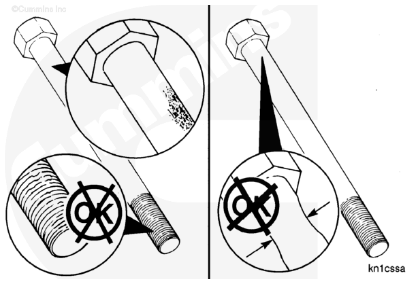
Note : If vacuum does not meet the specifications, the face of the valves and the valve seat inserts must be resurfaced. Refer to the Shop Manual, C Series Engines - Applicable to Engines Built in 1991 and After, Bulletin 3666008.



Cylinder Head Capscrews

Inspect the cylinder head capscrews for damaged threads, corroded surfaces, or a reduced diameter.

Note : A reduced diameter is caused by capscrew stretching.



Do **not** reuse cylinder head capscrews under the following conditions:

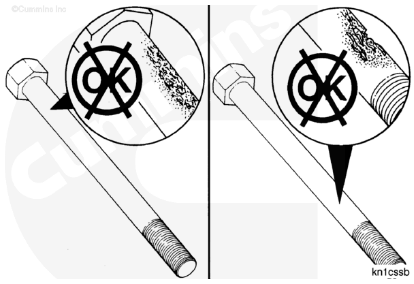
- Visible corrosion or pitting exceeds 1 cm^2 [0.155 in^2] in area.

Example:

- Acceptable - 9.53 mm [3/8 in] x 9.53 mm [3/8 in]
- Unacceptable - 12.7 mm [½ in] x 12.7 mm [½ in]

- Visible corrosion or pitting exceeds 0.12 mm [0.005 in] in depth.

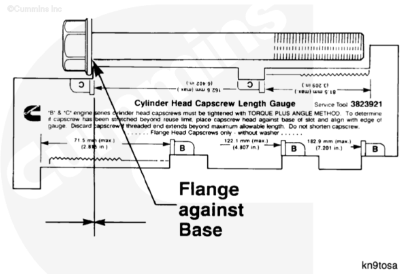
Note : Do not reuse a capscrew that has damaged threads or a reduced diameter.



- Visible corrosion or pitting that is located within 3.2 mm [0.126 in] of the fillet.
- Visible corrosion or pitting that is located within 3.2 mm [0.126 in] of the threads.
- Stretched beyond the “free-length” specifications.

Note : If the capscrews are not damaged, they can be reused throughout the life of the engine, unless the capscrew stretches beyond the “free-length” specification.

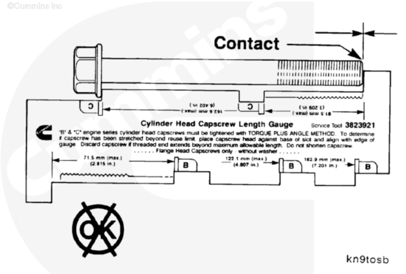
Use capscrew length gauge, Part Number 3823921, to measure the cylinder head capscrew.



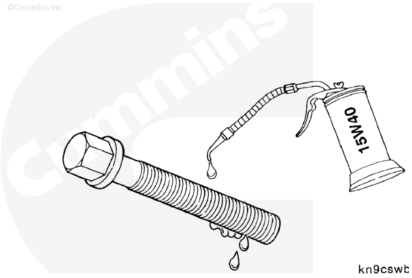
Cylinder Head Capscrew Free Length				
	mm		in	
Short Capscrews		81.5	MAX	3.2

Long Capscrews		162.6	MAX	6.4

Place the flange of the capscrew against the base of the slot on the capscrew length gauge. If the end of the capscrew touches the foot of the gauge, it is too long and the capscrew **must** be discarded.



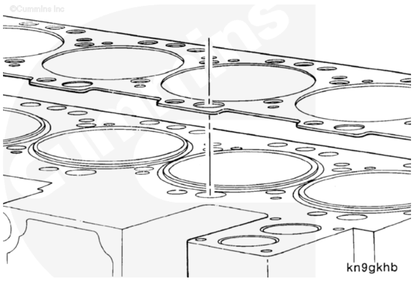
Immediately after cleaning and inspecting the capscrews, apply a light film of clean engine oil to all of the capscrews that are to be reused.



Install

⚠ CAUTION ⚠

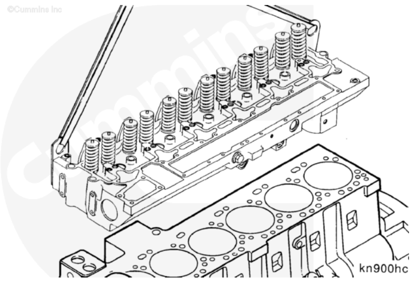
Make sure the cylinder head gasket is correctly aligned with holes in the cylinder block. If not aligned properly, it can cause engine damage.



Position the new cylinder head gasket over the dowels.

⚠ WARNING ⚠

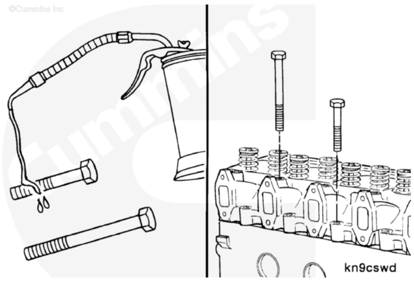
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Carefully put the cylinder head straight down onto the cylinder block, and seat it onto the dowels.

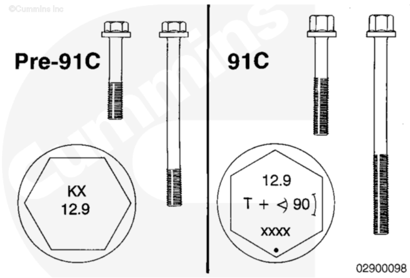
Lubricate the threads and under the heads of the remaining cylinder head capscrews with engine oil.

Install the capscrews hand-tight.



⚠ CAUTION ⚠

Do not use pre-1991 certification capscrews in a 1991 or later certification level engine because of differences in capscrew length and thread engagement. Damage to the engine will result if the wrong capscrews are used.



Post 1991 - New Cylinder Block or New Capscrews

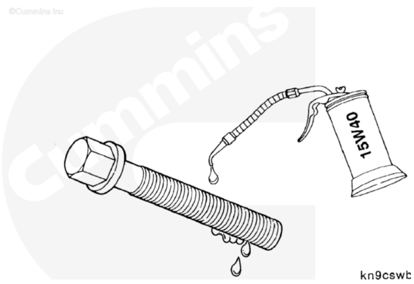
Always use the following procedure when you are using new capscrews or a new cylinder block.

Note : Each 1991 and later certification capscrew has been marked with symbols and an angle specification. The capscrew part numbers can be identified by inspecting the marks on the capscrew head.

Cylinder Head Capscrew Part Numbers			
Pre-91 Capscrew Part Number	Length	1991 Capscrew Part Number	Length
3907234	Short	3917729	Short
3907233	Long	3917728	Long

⚠ CAUTION ⚠

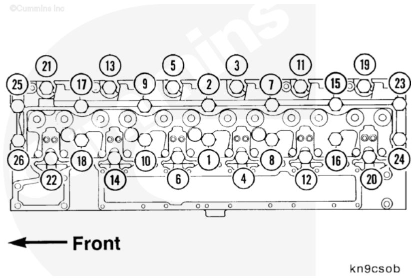
This procedure is to be used only with the use of a new cylinder block or new cylinder head capscrews.



1. Apply a thin film of clean engine lubricating oil to the capscrew threads and underside of the capscrew head flange.

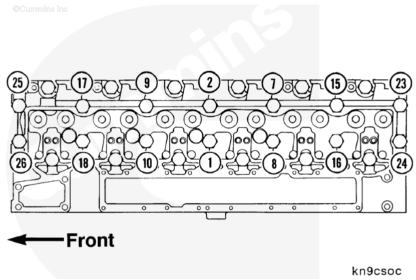
1. Follow the numbered torque sequence; tighten all capscrews.
2. Follow the numbered torque sequence; check the torque on all capscrews a second time.

Torque Value: 95 n•m [70 ft-lb]



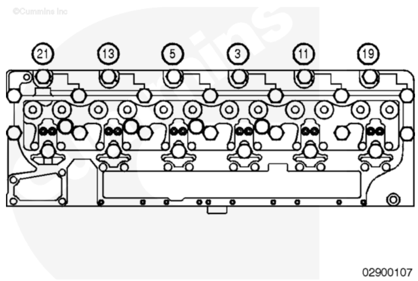
1. Follow the numbered sequence; tighten the 14 long capscrews (two center rows) **only**.
2. Follow the numbered sequence; check the torque on all 14 long capscrews a second time.

Torque Value: 145 n•m [107 ft-lb]



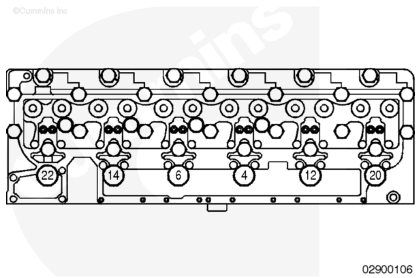
1. Follow the numbered sequence; tighten the six short capscrews (exhaust side) **only**.
2. Follow the numbered sequence; check the torque on all six short capscrews a second time.

Torque Value: 105 n•m [77 ft-lb]

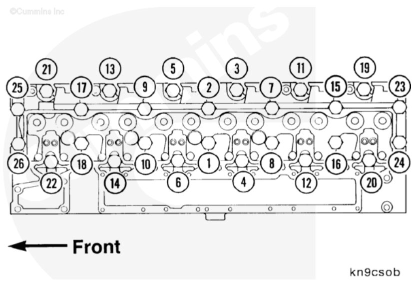


1. Follow the numbered sequence; tighten the six short capscrews (intake side) **only**.
2. Follow the numbered sequence; check the torque on all six short capscrews a second time.

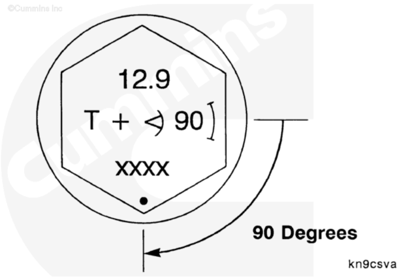
Torque Value: 95 n•m [70 ft-lb]



1. Recheck the torque on all capscrews.
2. Follow the numbered sequence; check the 14 long capscrews from step 4.
3. Follow the numbered sequence; check the 6 short capscrews (exhaust side) from step 6.
4. Follow the numbered sequence; check the 6 short capscrews (intake side) from step 8.

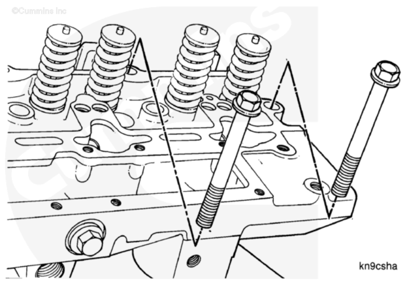


1. Follow the numbered sequence; from step 2, turn the capscrews 90 degrees as indicated on the capscrew head.

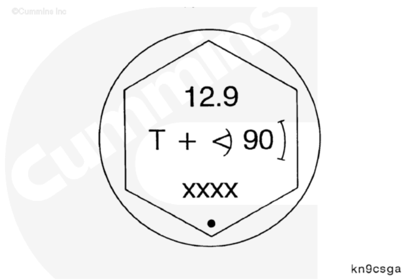


Post 1991 - Used Cylinder Block and Used Capscrews

Note : If using a new cylinder block or new capscrews, do not use this procedure. Reference the new cylinder block or new capscrews in this section.

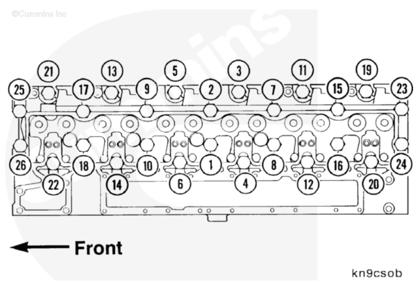


The top of the cylinder head capscrew is identified with an angle marking. The cylinder head capscrews **must** be tightened by the five-step “torque plus angle” method, as described here.



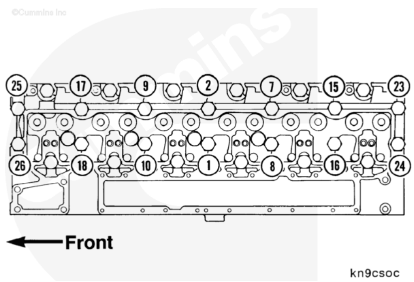
1. Follow the numbered sequence and tighten all 26 capscrews.

Torque Value: 70 n•m [52 ft-lb]



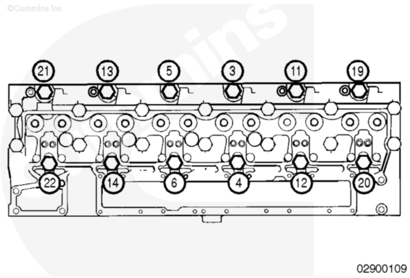
2. Follow the numbered sequence and tighten **only** the 14 long capscrews. (Number 1, 2, 7, 8, 9, 10, 15, 16, 17, 18, 23, 24, 25, and 26.)

Torque Value: 145 n•m [107 ft-lb]



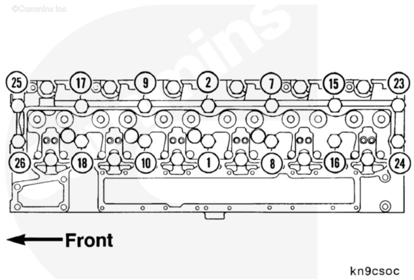
3. Retighten the short capscrews: Number 3, 4, 5, 6, 11, 12, 13, 14, 19, 20, 21, and 22 because of cylinder head relaxation and to obtain proper cylinder head torque requirements.

Torque Value: 70 n•m [52 ft-lb]

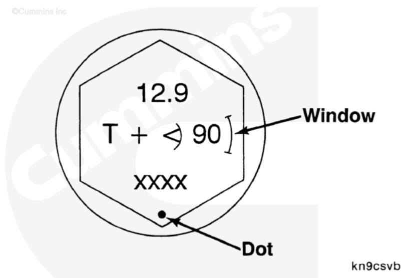


4. Follow the numbered sequence and retighten **only** the 14 long capscrews. (Number 1, 2, 7, 8, 9, 10, 15, 16, 17, 18, 23, 24, 25, and 26.)

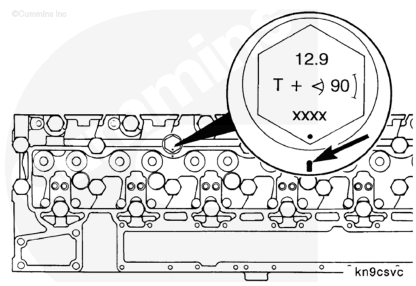
Torque Value: 145 n•m [107 ft-lb]



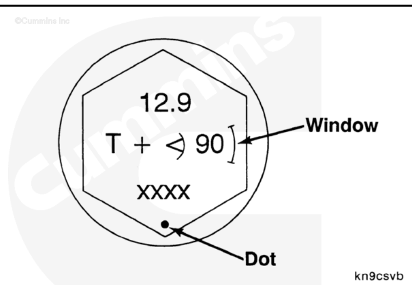
Note : To turn the capscrew accurately to the desired angle, orientate according to the small "dot" and "window" marked on the capscrew head, or use the torque angle gauge for ½-inch drive, Part Number 3823878, or torque angle gauge for ¾-inch drive, Part Number 3823879.



Mark the cylinder head adjacent to the dot on the capscrew head. This mark will serve as an indexing aid.



5. Rotate the capscrew until the mark that has been made on the cylinder head falls into the window on the capscrew head.

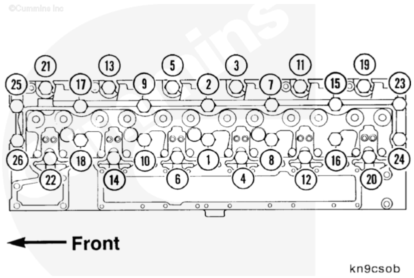


Pre - 1991 Certifications

1. Follow the numbered sequence and tighten all capscrews:

Torque Value:

1. 40 n•m [30 ft-lb]
2. 150 n•m [111 ft-lb]
3. 220 n•m [162 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the thermostat housing. Refer to Procedure 008-014 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-014.html)
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/41/41-011-007-tr.html)
- Install the rocker lever assemblies. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-008.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-014.html)
- Adjust the valve lash. Refer to Procedure 003-004 in Section 3. (/qs3/pubsys2/xml/en/procedures/41/41-003-004.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-011.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/41/41-

010-033-tr.html)

- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/41/41-006-026.html)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/41/41-006-051.html)
- Connect the coolant vent lines. Refer to Procedure 008-017 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-017.html)
- Install the fan pulley and fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/41/41-008-036.html)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine at idle 5 to 10 minutes to check for leaks and proper operation.

Last Modified: 07-Aug-2013